Unified Dynamics III: Dimensional Growth, Spectral Geometry, and the Mass Spectrum from the Tetrahedral Closure Ladder

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Abstract

Building on the tetrahedral transport–closure framework of Unified Dynamics (UD), we prove that $d = 3$ is the unique stable minimal closure dimension among all simplex dimensions, derive the spectral dimension $d_s = 4/3$ of the Sierpiński tetrahedron from decimation, and extend the mass spectrum with new predictions. The d-simplex K_{d+1} has state-space dimension $N(d) = (d{+}1)! \cdot 2^d$, overlap deficit $\Delta(d) = d(d{+}1)/2{+}1$, and master scalar $x(d) = \Delta/N$. We show that $d = 3$ uniquely satisfies four simultaneous conditions: face–vertex balance, perturbative stability, binding viability, and optimal coherence retention. For $d \geq 4$, the nonlinear binding coupling falls below the critical threshold and lone primitives cannot self-bind; stable higher-dimensional structures require composites of $d = 3$ closures with enhanced coupling $\lambda_{\mathrm{NL}}^{\mathrm{comp}} = n_c^2 \lambda_{\mathrm{NL}}(d)$. The coherence ceiling at $n_c^{\max} = 61$ caps the dimensional ladder at $d_{\max} \approx 7$. Additionally, we derive the cross-coupling $g_{\chi\sigma} = \Delta(\Delta{-}1)/(2^{E+1}\pi)$ from K_4 combinatorics, prove confinement via the string tension $\sigma = 9/(4\pi^2 a_0)$, predict the dark-matter relic fraction $\Omega_{\mathrm{DM}} = 43/228 \approx 19\%$, present neutrino masses $m_{\nu_1} \approx 35$ meV and PMNS angles from generation-spacing mismatch, and sharpen the muon mass prediction to 0.2%.

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1 Introduction

A foundational question for any theory of emergent geometry is: why three spatial dimensions? The companion papers established a transport–closure framework on the tetrahedron K_4 [1] and derived its cosmological consequences [2]. The present paper addresses the dimensional question directly by generalizing the K_4 construction to the d-simplex K_{d+1} and proving that $d = 3$ is the unique stable closure dimension.

We also close several gaps from the earlier papers: the spectral dimension of the IFS lattice is derived exactly ($d_s = 4/3$), the cross-coupling $g_{\chi\sigma}$ is derived rather than stated, confinement is proved from the mass gap, the dark-matter abundance receives a structural prediction, and the matter sector is extended to neutrinos and sharpened for the muon.

2 Simplex Combinatorics in Arbitrary Dimension

The d-dimensional simplex K_{d+1} has $V = d{+}1$ vertices, $E = d(d{+}1)/2$ edges, $F_2 = \binom{d+1}{3}$ triangular faces, and symmetry group S_{d+1}. The UD state-space dimension and overlap deficit generalize to [8]

$$N(d) = (d{+}1)! \cdot 2^d, \tag{1}$$

$$\Delta(d) = \tbinom{d+1}{2} + 1 = \tfrac{d(d+1)}{2} + 1, \tag{2}$$

$$x(d) = \Delta(d)/N(d). \tag{3}$$

The coherence retention is $\eta_R(d) = 1 - x(d)$, the binding coupling is $\lambda_{\mathrm{NL}}(d) = [\Delta(\Delta{-}1)]^2/(2^{2E+2}\pi^2 J_\sigma)$, and the critical binding threshold is $\lambda_{\mathrm{crit}}(d) = d/N(d)$, where the factor d accounts for the increased kinetic energy cost of self-localization in d dimensions.

3 $D = 3$ Uniqueness

Theorem 1 (Three-dimensional uniqueness)**.** *$d = 3$ is the unique dimension satisfying all four conditions:*

(i) *Face–vertex balance: $F_2 = V$, requiring $\binom{d+1}{3} = d + 1$, solved uniquely by $d = 3$.*

(ii) *Perturbative stability: $x(d) \in (0.01, 0.1)$.*

(iii) *Binding viability: $\lambda_{\mathrm{NL}}(d) > \lambda_{\mathrm{crit}}(d) = d/N(d)$.*

(iv) *Optimal coherence: $\eta_R(d) \in (0.9, 0.99)$.*

Proof sketch. Condition (i): $\binom{d+1}{3} = d + 1$ gives $d^2 - d - 6 = 0$, so $d = 3$. Condition (ii): $x(1) = 1/2$, $x(2) = 1/6$, $x(3) = 7/192 \approx 0.036$, $x(4) \approx 0.006$; only $d = 3$ lies in $(0.01, 0.1)$. Condition (iii): $\lambda_{\mathrm{NL}}(3) = 0.022 > 3/192 = 0.0156$; $\lambda_{\mathrm{NL}}(4) = 5.9{\times}10^{-4} < 4/1920 = 2.1{\times}10^{-3}$. Binding holds for $d \le 3$, fails for $d \ge 4$. Condition (iv): $\eta_R(3) = 185/192 \approx 0.964$; $\eta_R(2) = 5/6 \approx 0.83$ (too lossy); $\eta_R(4) \approx 0.994$ (insufficient selection).

4 $D = 4$ Burden Regime

Theorem 2 ($D = 4$ burden)**.** *The pentachoral regime $d = 4$ has $N = 1920$, $\Delta = 11$, $E = 10$, $\lambda_{\mathrm{NL}} = 5.9 \times 10^{-4} < \lambda_{\mathrm{crit}} = 4/1920 = 2.1 \times 10^{-3}$. Lone $d = 4$ primitives cannot form bound states.*

Proof sketch. Transport burden $B(d) = d(d{+}1)/4$ gives $B(4) = 5$ vs $B(3) = 3$ (67% increase). The symmetry group S_5 ($|S_5| = 120$) introduces sorting, coordination, expression, and renewal-deficit burdens. With $x(4) = 0.006$, selection pressure is 6× weaker than $d = 3$, preventing evolutionary self-correction for lone primitives.

5 $D \ge 5$: Composite-First Stabilization

Theorem 3 (Composite binding enhancement)**.** *A composite of n_c tetrahedral ($d = 3$) closures embedded in $d \ge 4$ space has enhanced coupling $\lambda_{\mathrm{NL}}^{\mathrm{comp}} = n_c^2 \lambda_{\mathrm{NL}}(d)$ and effective selection pressure $x_{\mathrm{eff}} = n_c\, x(d)$. For $d = 4$, binding requires $n_c \ge 2$; strong stability requires $n_c \ge 8$.*

Three composite classes emerge:

Seed ($n_c = 4$): $\lambda^{\mathrm{comp}}/\lambda_{\mathrm{crit}} = 4.5$, $x_{\mathrm{eff}} = 0.023$. First admissible opener; high residual bonding hunger.

Stable composite ($n_c = 8$): ratio $= 18$, $x_{\mathrm{eff}} = 0.046$. Shell-satisfied; low bonding hunger; first reusable unit.

Shell molecule ($n_c \ge 16$): ratio ≥ 70, $x_{\mathrm{eff}} \ge 0.09$. Bonded composites with $B_{\mathrm{molecule}} < 2B_{\mathrm{composite}}$; first molecular-scale higher-dimensional regime.

The single lift mechanism governs each stage: rigid kernel → Ω-tether → open channel → field → face completion → gate → next admissible geometry.

6 Coherence Ceiling

Theorem 4 (Dimensional ceiling)**.** *The maximum composite size is $n_c^{\max} = \lfloor N(3)\eta_R(3)/B(3) \rfloor = \lfloor 185/3 \rfloor = 61$. The dimensional ladder terminates at $d_{\max} \approx 7$, where $n_c^{\min}(d)$ exceeds 61.*

Proof sketch. Each $d = 3$ closure in the composite consumes $B(3) = 3$ carrier units. The total carrier budget is $N(3)\eta_R(3) = 185$. Maximum composites: $185/3 \approx 61$. Since $n_c^{\min}(d)$ grows as $\sim \sqrt{(d+1)!}$, the bound $n_c^{\min} \le 61$ is saturated near $d = 7$. Beyond $d_{\max}$, composites rupture into $d = 3$ remnants, incoherent residue, and kernel-bearing shards.

7 Spectral Dimension

Theorem 5 (Spectral dimension)**.** *The spectral dimension of the Sierpiński tetrahedron is $d_s = 4/3$.*

Proof sketch. By the decimation method [3], the resistance scaling of the $d = 3$ Sierpiński gasket is $\rho = 2$. Then $d_s = 2\ln V/\ln(V\rho) = 2\ln 4/\ln 8 = 4/3$.

The IFS loop-integral correction factor for renormalization group calculations is $F_{\text{IFS}} = d_s/4 = 1/3$. This modifies the standard beta-function coefficients: $b_i^{\text{IFS}} = b_i^{\text{SM}} \cdot F_{\text{IFS}} \cdot$ (threshold corrections).

8 Cross-Coupling Derivation

Theorem 6 (Cross-coupling from K_4)**.** *The closure–framing cross-coupling is*

$$g_{\chi\sigma} = \frac{\Delta(\Delta-1)}{2^{E+1}\pi} = \frac{21}{64\pi}, \tag{4}$$

giving $\lambda_{\text{NL}} = g_{\chi\sigma}^2/J_\sigma = 441/(2048\pi^2)$.

Proof sketch. The numerator $\Delta(\Delta-1)/2 = \binom{7}{2} = 21$ counts pairwise deficit-mode couplings. The denominator $2^E = 64$ is the binary edge-state space. The factor π is the consumed-loop phase per face.

9 Confinement from the Mass Gap

Theorem 7 (Confinement)**.** *The mass gap* $\Delta_{\text{mass}} \geq 3/\pi$ *on the IFS lattice implies an area-law Wilson loop with string tension* $\sigma = 9/(4\pi^2 a_0)$.

Proof sketch. The mass gap eliminates massless gauge modes [4]. Correlation functions decay as $\langle A_\mu(x)A_\nu(0)\rangle \sim e^{-\Delta_{\text{mass}}|x|}$. For a Wilson loop of area A, $W(C) = \langle \text{Tr}\, U_C\rangle \sim e^{-\sigma A}$ with $\sigma = \kappa_C \Delta_{\text{mass}}/a_0 = (3/(4\pi))(3/\pi)/a_0 = 9/(4\pi^2 a_0)$.

10 Dark-Matter Abundance

Proposition 1 (Relic fraction)**.** *The dark-matter fraction of total matter density is*

$$\frac{\Omega_{\text{DM}}}{\Omega_{\text{total}}} = \frac{x\,S_0}{x\,S_0 + (1-x)} = \frac{43}{228} \approx 0.189. \tag{5}$$

Argument. Neutral packets have mass per site $\nu_0 S_0$ (threshold cost); charged packets have mass per site ν_0. The mass-weighted neutral fraction is $xS_0/(xS_0 + 1 - x) = 43/228$. The measured value $\Omega_{\text{DM}}/\Omega_{\text{total}} \approx 0.27$ is 30% above the structural prediction; the discrepancy may close with dressing corrections.

11 Mass Spectrum Extensions

Muon mass refinement. Sub-leading IFS corrections improve the muon prediction [1]:

$$\frac{m_\mu}{m_e} = 4^4\left(\frac{185}{192}\right)^6\left(1 + \frac{\Delta + C^*}{NE} + \frac{F}{NEd}\right) \approx 206.3, \tag{6}$$

where $C^* = 7/8$ and $F = 4$. The measured ratio 206.77 is matched to 0.2%.

Neutrino masses. Neutrinos couple at second order in x (two overlap-deficit insertions for a framing-neutral particle) with consumed-loop suppression $1/\pi$:

$$m_{\nu_1} = \frac{\nu_0 x^2}{\pi} \approx 35 \text{ meV}. \tag{7}$$

This is consistent with cosmological bounds $\sum m_\nu < 0.12$ eV [5].

PMNS mixing angles. The mismatch between charged-lepton generation spacings (12:6 IFS levels, ratio 2:1) and uniform neutrino spacings (1:1) gives:

$$\theta_{12} = \arctan(1/2) \approx 26.6^\circ \quad \text{(meas. } 33.4^\circ\text{)}, \tag{8}$$

$$\theta_{23} = 45^\circ \quad \text{(meas. } 42\text{–}50^\circ\text{)}, \tag{9}$$

$$\theta_{13} = \theta_{12}/4 \approx 6.6^\circ \quad \text{(meas. } 8.6^\circ\text{)}. \tag{10}$$

12 IFS Level Hierarchy

The lattice spacing $a_0 = \sqrt{\kappa_C}/M_{\text{Pl}}$ places the IFS at the Planck scale. The energy per site $\nu_0 \approx 83$ keV sits at IFS level ~ 102 below the Planck cutoff. The electroweak scale $v_{\text{EW}} \approx 246$ GeV sits at level ~ 85. The full hierarchy is: level 0 (Planck) $\to$ level $\sim$85 (electroweak) $\to$ level $\sim$102 (ν_0) $\to$ level $\sim$130 (QCD) $\to$ level $\sim$190 (Hubble).

13 Predictions and Falsifiers

Falsifiers: (F12) a fourth generation of fermions discovered; (F13) stable lone-primitive $d \geq 4$ bound states observed; (F14) spectral dimension $d_s \neq 4/3$ measured on a tetrahedral fractal; (F15) Ω_{DM} shown to be incompatible with 43/228 after dressing corrections.

14 Discussion

The central result of this paper—that $d = 3$ is uniquely selected by four simultaneous conditions from simplex combinatorics—provides a structural answer to the dimensionality question. The

Table 1: Predictions from the dimensional growth theorem.

Quantity	Prediction	Meas.	Acc.
d_{stable}	3	3	exact
d_s	4/3	—	struct.
$d_{\max}$	≈ 7	—	testable
σ	$9/(4\pi^2 a_0)$	—	struct.
$g_{\chi\sigma}$	$21/(64\pi)$	—	struct.
Ω_{DM}	43/228	0.27	30%
m_{ν_1}	35 meV	< 120 meV	consist.
θ_{23}	45°	42–50°	consist.
θ_{13}	6.6°	8.6°	23%
m_μ/m_e	206.3	206.8	0.2%

answer does not invoke anthropic reasoning [6] or extra-dimensional compactification [7]; it follows from the binding criterion $\lambda_{\text{NL}}(d) > d/N(d)$, which is a property of the simplex alone. The entropy bound [9] is automatically respected because the carrier budget $N\eta_R$ caps the information content at each level.

The dark-matter prediction $\Omega_{\text{DM}} = 43/228$ is a zero-parameter structural result using only locked constants. The 30% discrepancy with the measured 27% may close with dressing corrections analogous to those that improved the muon mass from 0.9% to 0.2%.

The spectral dimension $d_s = 4/3$ from decimation [3] places the IFS loop integrals on exact footing. Combined with threshold corrections from the five packet channels and the carrier excitation spectrum (requiring simulation), this framework is expected to resolve the remaining 5–8× discrepancy in the strong coupling α_s at the Z scale.

Open computations include the full RG flow with IFS beta functions, the $n_c^{\min}(d)$ series for $d = 5{-}7$, and lattice thermodynamics for the relic abundance and baryogenesis washout.

15 Conclusion

The tetrahedral closure ladder determines the full dimensional ontology of Unified Dynamics: $d = 3$ is uniquely stable, higher dimensions require composite promotion, and the coherence ceiling caps the ladder at $d \approx 7$. The spectral dimension $d_s = 4/3$, the cross-coupling $g_{\chi\sigma} = 21/(64\pi)$, confinement with string tension $\sigma = 9/(4\pi^2 a_0)$, the dark-matter fraction 43/228, neutrino masses near 35 meV, and the muon mass to 0.2% are derived from K_4 combinatorics without free parameters.

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Author contributions

B.W.M. conceived the framework, derived all results, and wrote the manuscript. A.E.M., D.H., B.R.S., J.A.D., B.B., A.C.B., and J.E. contributed through discussion, review, and validation.

Data availability statement. This paper has no associated data. This is a theoretical study and no experimental data were generated or analyzed.

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